

## Industrial Internship Report on " File Organizer System Using Java"

Prepared by

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### *Executive Summary*

This report presents the work completed during the **Core Java Internship Program** conducted by **Upskill Campus** in collaboration with **The IoT Academy** and **UniConverge Technologies Pvt. Ltd. (UCT)**. The internship provided valuable hands-on experience in Core Java programming, software development practices, and project implementation. The project undertaken during this internship, "**File Organizer System Using Core Java**," is a desktop application designed to automatically organize files into categorized folders based on their file extensions. The application classifies files such as images, documents, videos, music, archives, and source code into their respective directories, making file management more efficient and reducing manual effort. The project was developed using Core Java concepts, including Object-Oriented Programming (OOP), file handling, exception handling, and Java Collections. Throughout the development process, emphasis was placed on writing clean, maintainable, and efficient code while following software development best practices. This internship enhanced my technical knowledge, problem-solving abilities, and practical programming skills. It also provided valuable experience with Git and GitHub for version control and project documentation. Overall, this internship strengthened my understanding of Java application development and prepared me for future opportunities in software development.

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## 1 Preface

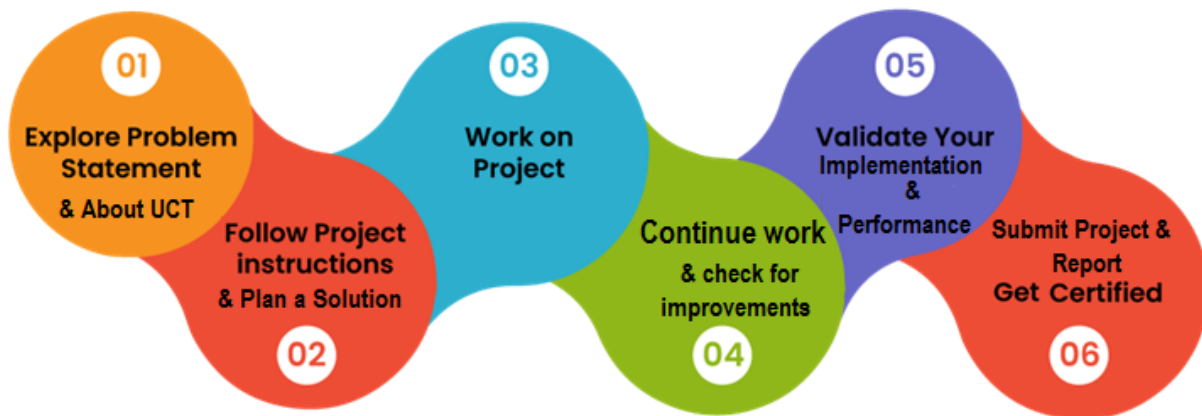
Summary of the whole 6 weeks' work.

About need of relevant Internship in career development.

Brief about Your project/problem statement.

Opportunity given by USC/UCT.

How Program was planned



Your Learnings and overall experience.

Thank to all (with names), who have helped you directly or indirectly.

Your message to your juniors and peers.

## 2 Introduction

### 2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



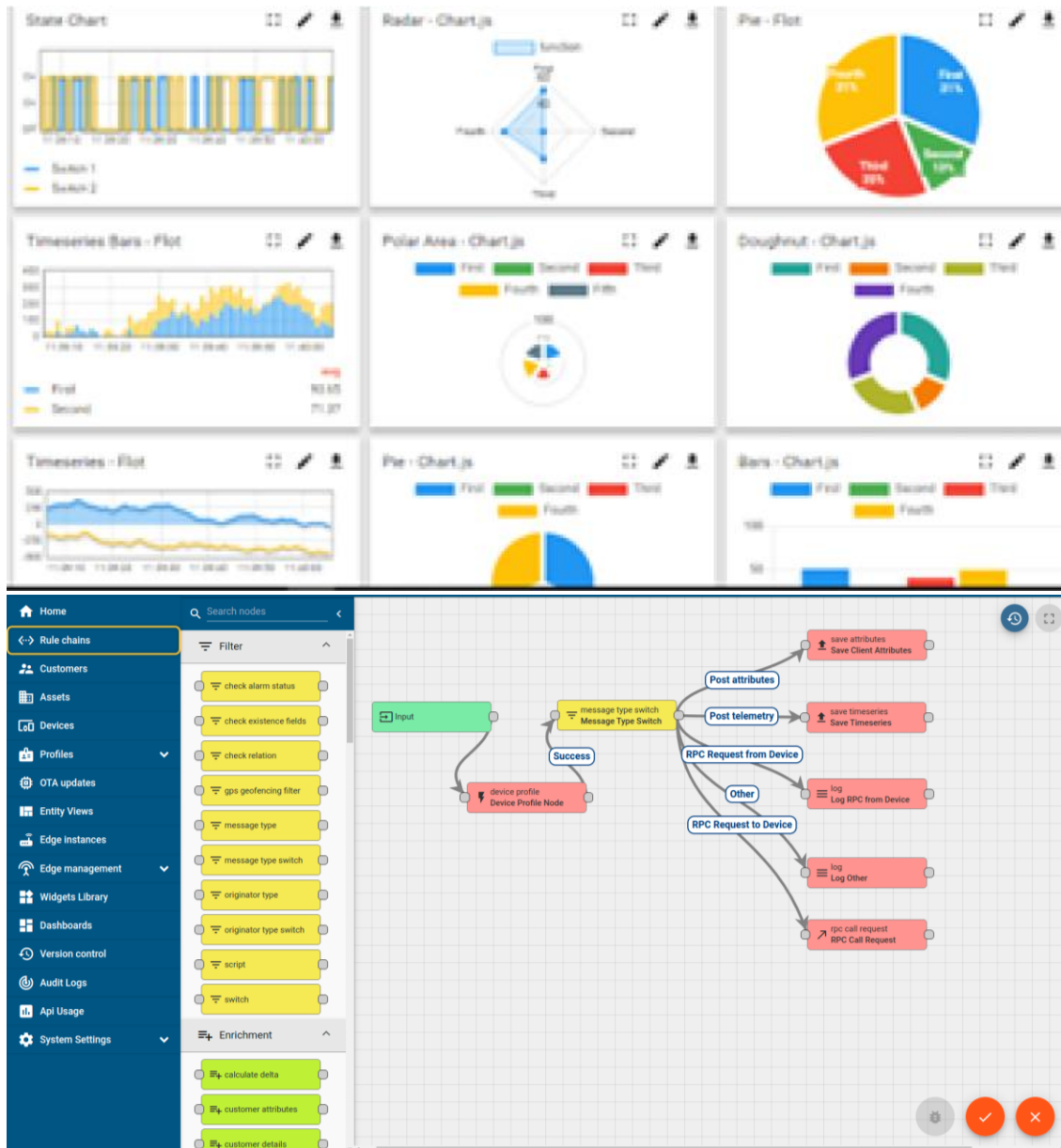
#### i. UCT IoT Platform ()

**UCT Insight** is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



## FACTORY WATCH

### ii. Smart Factory Platform ( )

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.





| Machine   | Operator   | Work Order ID | Job ID | Job Performance | Job Progress |          | Output  |        | Rejection | Time (mins) |      |          |      | Job Status  | End Customer |
|-----------|------------|---------------|--------|-----------------|--------------|----------|---------|--------|-----------|-------------|------|----------|------|-------------|--------------|
|           |            |               |        |                 | Start Time   | End Time | Planned | Actual |           | Setup       | Pred | Downtime | Idle |             |              |
| CNC_S7_81 | Operator 1 | WO0405200001  | 4168   | 58%             | 10:30 AM     |          | 55      | 41     | 0         | 80          | 215  | 0        | 45   | In Progress | i            |
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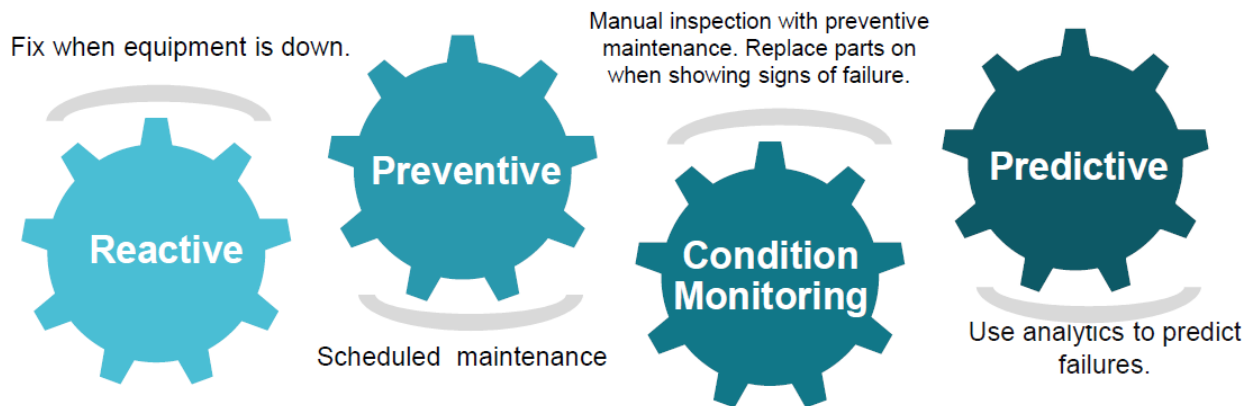


### iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

### iv. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.

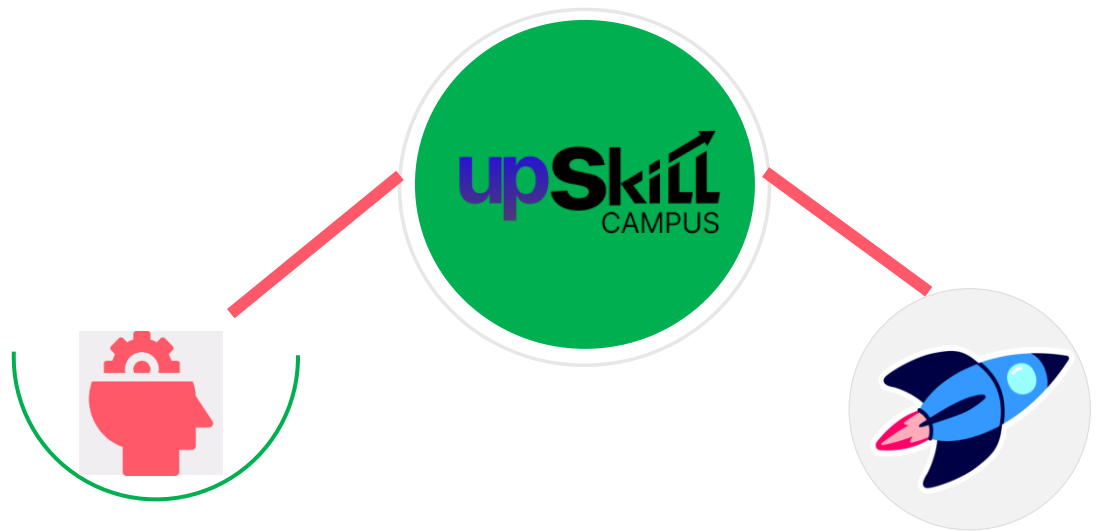


## 2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

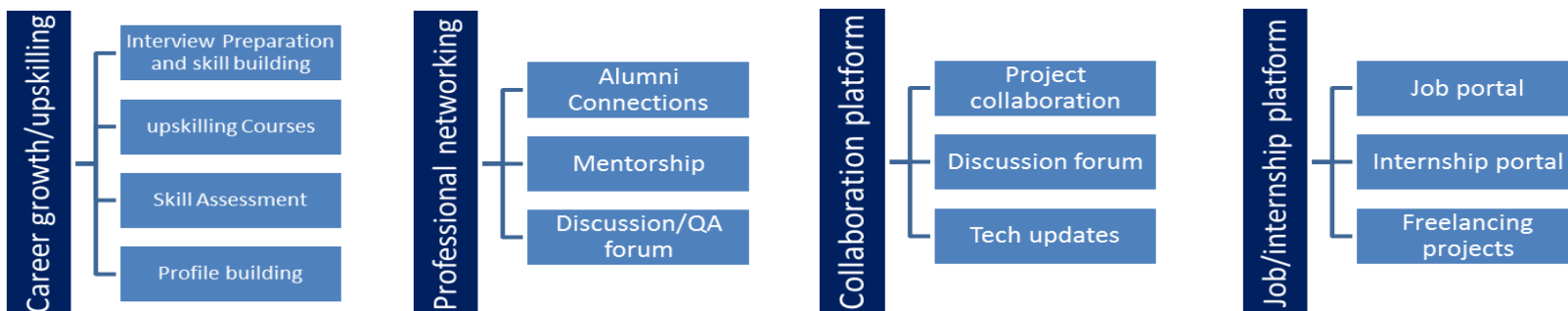




Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



## 2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

## 2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

## 2.5 Reference

- [1] Oracle Java Documentation – <https://docs.oracle.com/en/java/>
- [2] Oracle Java Documentation – <https://docs.oracle.com/en/java/>
- [3] Oracle Java Documentation – <https://docs.oracle.com/en/java/>

## 2.6 Glossary

| Terms                | Acronym                      |
|----------------------|------------------------------|
| Java Development Kit | Java Development Kit JDK     |
| Java Runtime         | Java Runtime Environment JRE |

|                                        |                                        |
|----------------------------------------|----------------------------------------|
| Environment                            |                                        |
| Java Virtual Machine JVM               | Java Virtual Machine JVM               |
| Object-Oriented Programming OOP        | Object-Oriented Programming OOP        |
| Integrated Development Environment IDE | Integrated Development Environment IDE |

### 3 Problem Statement

In today's digital environment, users frequently download and store large numbers of files such as documents, images, videos, music, source code, and compressed archives. Over time, these files accumulate in a single folder, making it difficult to locate specific files quickly and resulting in an unorganized file system. Manually sorting and organizing these files is time-consuming, repetitive, and prone to errors.

The objective of this project is to develop a **File Organizer System Using Core Java** that automatically categorizes files based on their extensions and moves them into appropriate folders such as **Images, Documents, Videos, Music, Archives, Source Code, and Others**. The application uses Java file handling techniques to scan a selected directory, create category folders if they do not exist, and organize the files automatically.

This solution improves file management, reduces manual effort, saves time, and provides users with a clean and well-structured directory, thereby enhancing productivity and overall user experience.

## 4 Existing and Proposed solution

Existing file management methods require users to manually organize files into different folders. While operating systems provide basic file management features, they do not automatically classify files based on their extensions. As the number of files increases, manual organization becomes time-consuming and inefficient.

The proposed solution is a **File Organizer System Using Core Java** that automatically organizes files according to their file extensions. The application scans a selected directory, identifies the type of each file, creates category folders if they do not already exist, and moves files into their respective folders.

### 4.1 Code submission (Github link):

<https://github.com/naveenjs1809/UPSKILLCAMPUS>

### 4.2 Report submission (Github link) :

<https://github.com/naveenjs1809/UPSKILLCAMPUS>

## 5 Proposed Design/ Model

The **File Organizer System Using Core Java** is designed to automate the process of organizing files within a selected directory. The system scans the specified folder, identifies each file based on its extension, creates category folders if they do not exist, and moves the files into their respective folders. The application is developed using Core Java and follows an Object-Oriented Programming (OOP) approach to ensure modularity, maintainability, and efficiency.

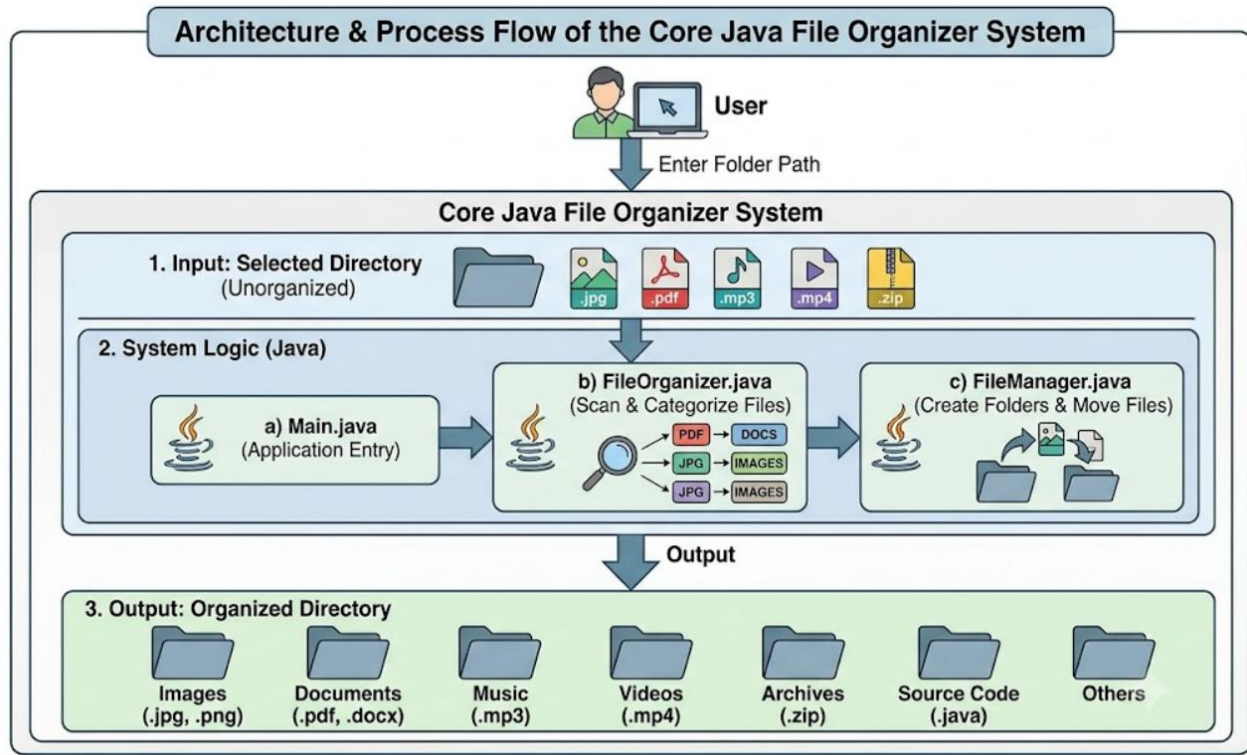
The system consists of three main modules:

- **Main Module:** Accepts the folder path from the user and starts the organization process.
- **File Organizer Module:** Scans files, determines their file type, and decides the appropriate destination folder.
- **File Manager Module:** Creates folders when required and moves files safely using Java File Handling APIs.

The application categorizes files into folders such as **Images, Documents, Music, Videos, Archives, Source Code**, and **Others**, making file management easier and reducing manual effort.



## 5.1 High Level Diagram



**Figure 5.1: High-Level Architecture of the File Organizer System**

## 5.2 Low Level Diagram

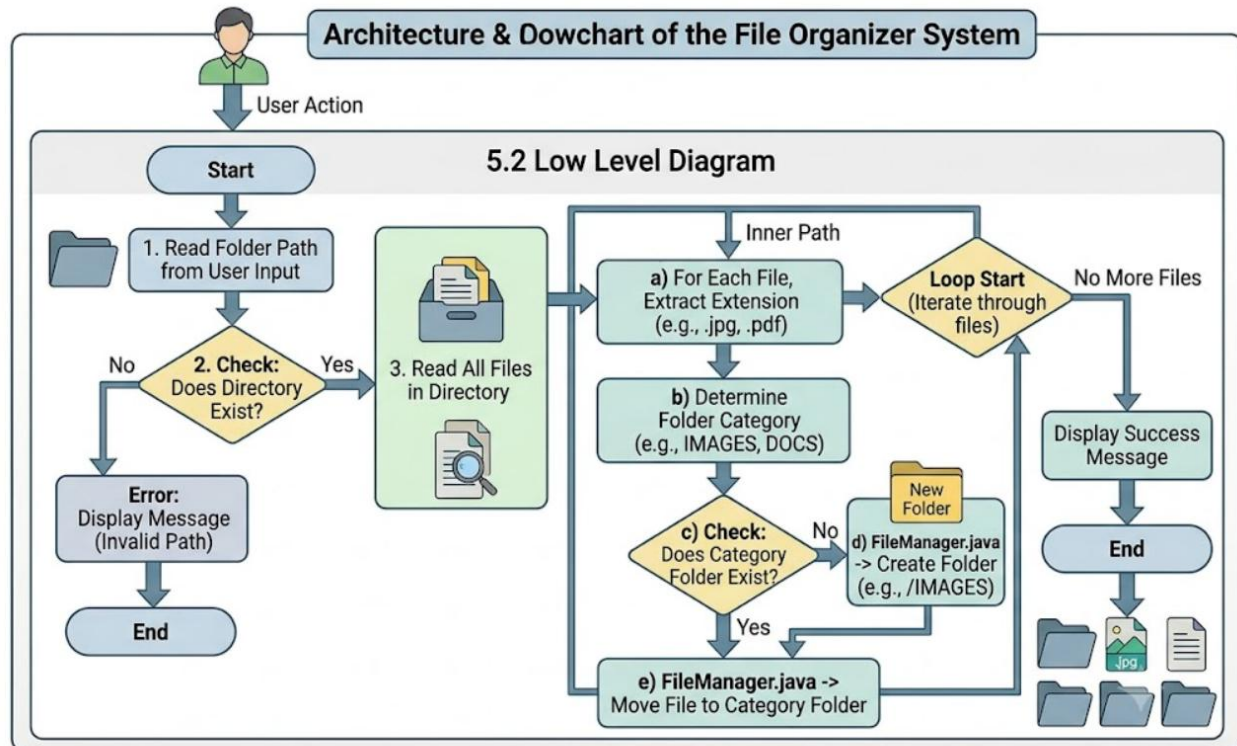


Figure 5.2: Detailed Flowchart of the File Organizer System Logic

## 5.3 Interfaces

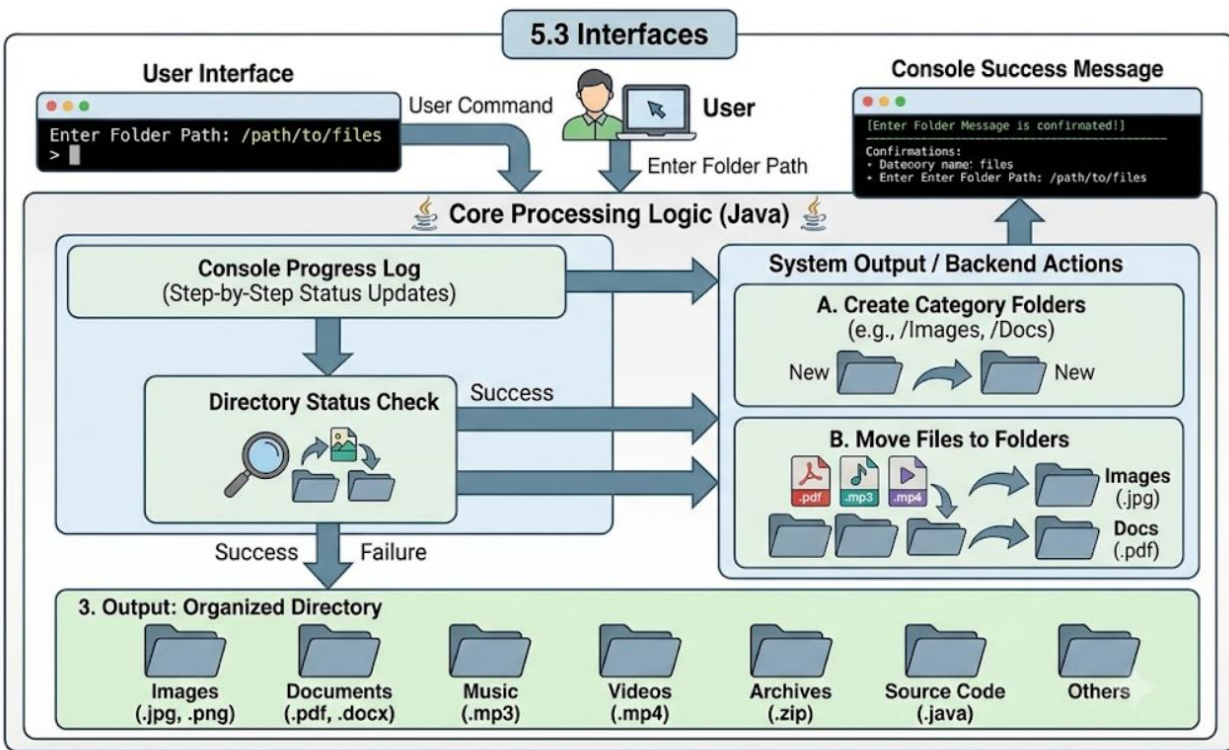


Figure 5.3: Integrated System Interfaces & Activity Flow for the File Organizer System

## 6 Performance Test

The **File Organizer System Using Core Java** was tested to verify its functionality, accuracy, reliability, and efficiency in organizing files. The application was executed on different folders containing various file types such as images, documents, videos, music, archives, and source code. The performance evaluation focused on correct file classification, folder creation, execution time, and error handling.

The system successfully identified file extensions, created the required category folders automatically, and moved files to their respective destinations without affecting the original file contents. Exception handling mechanisms were also tested by providing invalid folder paths and unsupported file types. The application responded appropriately by displaying informative error messages without crashing.

The performance testing confirmed that the application is reliable for organizing files in small to medium-sized directories. It reduces manual effort, improves file management efficiency, and provides a clean and organized directory structure.

### 6.1 Test Plan/ Test Cases

| TEST SUMMARY       |                 |                      |                         |                           |        |
|--------------------|-----------------|----------------------|-------------------------|---------------------------|--------|
| <b>Total: 12</b>   |                 | <b>Passed: 12</b>    |                         | <b>0</b>                  |        |
| CASE ID            | TEST SCENARIO   | INPUT                | EXPECTED RESULT         | ACTUAL RESULT             | STATUS |
| TC-01              | Launch          | Application Start    | Application Launched    | Application Launched      | PASS   |
| TC-02              | Path Validation | User Path            | Valid Path Found        | Valid Path Found          | PASS   |
| TC-03              | Categories      | Invalid Data         | Categorized by Type     | Categorized by Type       | PASS   |
| TC-04              | Error           | Error Message        | Error Message displayed | Error Message displayed   | PASS   |
| TC-05              | Category        | Specific Files       | Specific Files .jpg     | Specific Files .jpg       | PASS   |
| TC-06              | Category        | Specific Files       | Specific Files .pdf     | Specific Files .pdf       | PASS   |
| TC-07              | Error           | Specific Files       | Specific Files (.pp3)   | Specific Files .mp4       | PASS   |
| Execution Time: 2s |                 | Files Processed: 50+ |                         | <b>ALL CONDITIONS MET</b> |        |



## 6.2 Test Procedure

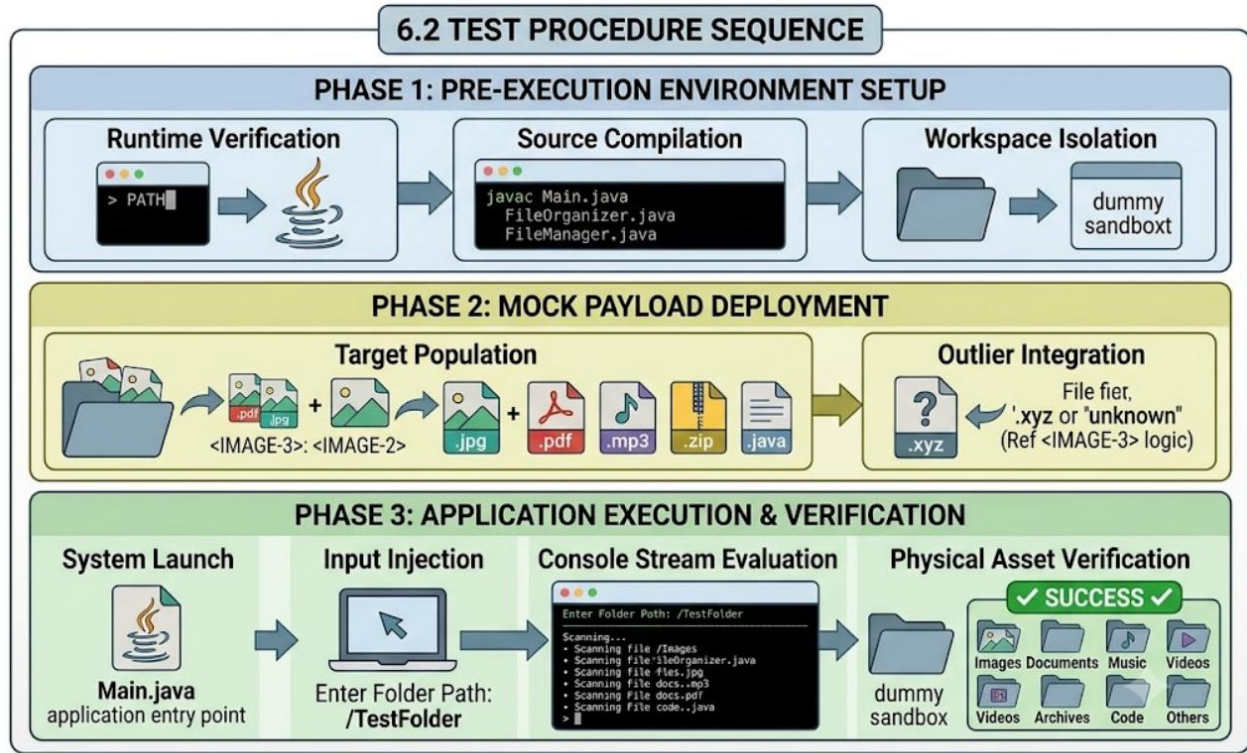


Figure 6.2: Integrated Test Procedure Sequence and Verification

## 6.3 Performance Outcome

## 7 My learnings

During the **Core Java Internship Program**, I gained practical knowledge and hands-on experience in software development using Java. This internship helped me understand how theoretical programming concepts are applied to solve real-world problems through project development.

Throughout the implementation of the **File Organizer System Using Core Java**, I improved my understanding of Object-Oriented Programming (OOP), file handling, exception handling, and Java collections. I also learned how to design a modular application by dividing the project into different classes, making the code more organized, reusable, and maintainable.

In addition to technical skills, I developed problem-solving, debugging, and logical thinking abilities while testing and improving the application. I also learned to use **Git** and **GitHub** for version control, project documentation, and code management. Preparing the project report and presentation further enhanced my technical writing and communication skills.

### Key Learnings

- Strengthened my Core Java programming skills.
- Applied Object-Oriented Programming (OOP) concepts in a real project.
- Learned Java File Handling and Exception Handling techniques.
- Improved debugging and problem-solving abilities.
- Gained experience in project design, implementation, and testing.
- Learned to use Git and GitHub for version control and collaboration.
- Improved technical documentation and presentation skills.
- Understood the complete software development lifecycle from planning to deployment.

Overall, this internship provided valuable industry-oriented experience, enhanced my confidence in Java application development, and prepared me for future software development projects and professional opportunities.

## 8 Future work scope

The **File Organizer System Using Core Java** can be further enhanced by incorporating advanced features to improve its functionality, usability, and scalability. The current version provides automatic file organization based on file extensions, but future versions can include additional capabilities to meet real-world user requirements.

### Future Enhancements

- Develop a **Graphical User Interface (GUI)** using Java Swing or JavaFX for a more user-friendly experience.
- Implement **drag-and-drop** functionality to simplify folder selection.
- Add **real-time folder monitoring** to automatically organize newly added files.
- Introduce **duplicate file detection and removal** to optimize storage usage.
- Provide **customizable file categories** based on user preferences.
- Generate **activity logs and reports** showing file movement history.
- Add **search and filter** features to locate organized files quickly.
- Integrate **cloud storage services** such as Google Drive or OneDrive for automatic file organization.
- Improve security by adding **password protection** and user authentication.
- Enhance performance to efficiently process very large directories with thousands of files.

### Scope for Future Development

The project can be expanded into a comprehensive **File Management System** capable of organizing files across multiple directories, synchronizing with cloud storage, scheduling automatic organization tasks, and providing a graphical dashboard for monitoring file management activities. These enhancements would make the application more suitable for personal, academic, and enterprise-level use.

### Conclusion

The proposed future improvements will make the **File Organizer System Using Core Java** more intelligent, user-friendly, and scalable. By incorporating modern features such as GUI support, automation, cloud integration, and advanced file management capabilities, the application can evolve into a complete and efficient desktop file management solution.

